P1.7 Energy and efficiency

Learning objectives

After this topic, you should know:

- what is meant by efficiency
- what is the maximum efficiency of any energy transfer
- how machines waste energy
- (H) how energy transfers can be made more efficient.

When you lift an object, energy stored in your muscles is transferred to the gravitational potential energy store of the object. The amount transferred depends on the object's weight and how high you lift it.

- Weight is measured in newtons (N). The weight of a 1 kilogram object on the Earth's surface is about 10 N.
- Energy is measured in joules (J). The energy needed to lift a weight of 1 newton by a height of 1 metre is equal to 1 joule.

The energy supplied to the device is called the input energy. The useful energy transferred by the device is called the useful output energy.

Because energy cannot be created or destroyed:

Input energy (energy supplied, J) = useful output energy (useful energy transferred, J) + energy wasted (J)

For any device that transfers energy:

$$\textbf{efficiency} = \frac{\textbf{useful output energy transferred by the device (J)}}{\textbf{total input energy supplied to the device (J)}}$$

Efficiency

Efficiency can be written as a decimal number (that is always less than 1) or as a percentage.

For example, a light bulb with an efficiency of 0.15 would radiate 15 J of energy as light for every 100 J of energy you supply to it.

- Its efficiency (as a number) $= \frac{15}{100} = 0.15$
- Its percentage efficiency $= 0.15 \times 100\% = \mathbf{15\%}$

Worked example

An electric motor is used to raise an object. The object's gravitational potential energy store increases by 60 J when the motor is supplied with 200 J of energy by an electric current. Calculate the percentage efficiency of the motor.

Solution

Total energy supplied to the device = 200 J

Useful energy transferred by the device = 60 J

Percentage efficiency of the motor

$$= \frac{\text{useful output energy transferred by the device}}{\text{total input energy supplied to the device}} \times 100\%$$

$$= \frac{60\text{ J}}{200\text{ J}} \times 100\% = 0.30 \times 100\% = \mathbf{30\%}$$

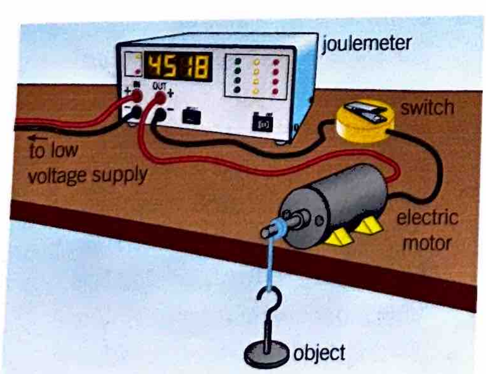

Figure 1 *An electric winch*

Efficiency limits

No device can be more than 100% efficient, because you can never get more energy from a machine than you put into it.

Investigating efficiency

Figure 1 shows how you can use an electric winch to raise a weight. You can use the joulemeter to measure the energy supplied.

- If you double the weight for the same increase in height, do you need to supply twice as much energy to do this task?

To calculate the change in the gravitational potential energy store of the weight use the following equation:

gravitational potential energy = weight in newtons × height increase in metres.

Use this equation and the joulemeter measurements to work out the percentage efficiency of the winch.

- Determine how the efficiency depends on the weight of the object being raised.

Plot a graph of your results, and use it to discuss how the efficiency of the winch changes with weight.

- Make some more measurements to find out whether lubricating the axle of the electric motor with a few drops of suitable oil makes the winch more efficient. Switch the motor off when you lubricate it.

Safety: Protect the floor and your feet. Stop the winch before the masses wrap round the pulley.

Study tip

The greater the percentage of energy that is usefully transferred in a device, the more efficient the device is.

Higher

Improving efficiency

Table 1 *Increasing the efficiency of devices*

	Why devices waste energy	How to reduce the problem
1	Friction between the moving parts causes heating.	Lubricate the moving parts to reduce friction.
2	The resistance of a wire causes the wire to get hot when a current passes through it.	In circuits, use wires with as little electrical resistance as possible.
3	Air resistance causes a force on a moving object that opposes its motion. Energy transferred from the object to the surroundings by this force is wasted.	Streamline the shapes of moving objects to reduce air resistance.
4	Sound created by machinery causes energy transfer to the surroundings.	Cut out noise (e.g., tighten loose parts to reduce vibration).

Study tip

Efficiency and percentage efficiency are numbers without units. The maximum efficiency is 1, or 100%, so if your calculation produces a number greater than this, it must be wrong.

1 **a** Compare the useful output energy by a machine and the input energy supplied to the machine. [1 mark]
b Explain why the percentage efficiency of a machine can never be:
i more than 100% [2 marks] **ii** equal to 100%. [4 marks]

2 An electric motor is used to raise a weight. When you supply 60 J of energy to the motor, the weight gains 24 J of gravitational potential energy. Calculate:
a the energy wasted by the motor [1 mark]
b the efficiency of the motor. [2 marks]

3 A machine is 25% efficient. If the total energy supplied to the machine is 3200 J, calculate how much useful energy can be transferred. [2 marks]

4 An electric fan heater contains a motor that blows hot air into the room. Describe the changes in the energy stores of the heater. [4 marks]

Key points

- The efficiency of a device = useful energy transferred by the device ÷ total energy supplied to the device (× 100%).
- No energy transfer can be more than 100% efficient.
- Machines waste energy because of friction between their moving parts, air resistance, electrical resistance, and noise.
- H Machines can be made more efficient by reducing the energy they waste. For example, lubrication is used to reduce friction between moving parts.